Eric Forbes

5G Systems Integration Lead · Lockheed Martin · PE, PMP, PhD Candidate
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PROFILE

Wireless product leader grown from bench-level test engineering to directing O-RAN deployments, sub-contractors, and customer integrations. I own products from the top down — architecture, technical direction, customers — with hands-on depth at every layer beneath. I build teams that make products, investing in long-term strategy and team culture over short-term wins.

EXPERIENCE

Lockheed Martin *May 2021 – Present*

5G Systems I&T Lead, RMS *November 2024 – Present*

Product owner for O-RAN and Tactical MANET deployments connecting mission-critical applications.

- Set technical direction for O-RAN software deployment on low-SWaP platforms; direct and coordinate sub-contractor teams.
- Own system architecture models through to hands-on, on-site integration for internal and external customers.

5G Systems I&T Engineer, RMS *April 2024 – November 2024*

Demonstrated the world's first 5G VPX CMOSS system with open-source software.

- Executed L2/L3 network integration, PTP timing analysis, 5G RAN/Core log debugging, and end-to-end system testing.

4G Systems Engineer, Space *August 2022 – April 2024*

Brought the first LTE GEO Satellite system to close.

- Performed full-stack eNB analysis (UE modem & eNB); developed test plans and authored program documentation.
- Applied Python for large dataset analysis; Bash scripting for system debugging and automation.

5G Systems Engineer, Space *May 2021 – August 2022*

Defined architecture and requirements for 5G Non-Terrestrial Networks (NTN).

- Developed NTN requirements and network diagrams; tracked 3GPP Rel-17 NTN working groups to position the product for where the standard was heading.

L3Harris Technologies *February 2020 – May 2021*

Sr. Specialist, Systems Engineer

Requirements Lead for a new Radar Warning Receiver (RWR) System program.

- Owned requirements development and decomposition in IBM DOORS; engaged prime customers for requirements review and program analysis.
- Applied MagicDraw & SysML for MBSE across the Space and Airborne Systems division.

G3 Technologies, Inc. *January 2017 – February 2020*

Cellular Systems Test Engineer

Protocol testing across four generations of wireless standards for commercial and defense customers.

- Debugged and tested per 3GPP/3GPP2 standards (LTE, UMTS, GSM, CDMA); analyzed Qualcomm chipsets (QXDM).
- Performed physical layer (RF) testing and verification; authored full system manuals and customer documentation.

Intertek *January 2015 – January 2017*

Project Engineer

Led compliance testing and certification for consumer electronics to ANSI, CA, and IEC safety standards.

- Managed customer timelines, executed test plans, and delivered compliance reports for end-product certification.
- Served as Safety Officer; completed OSHA 40-hour training.

EDUCATION

Stevens Institute of Technology *2020 – Fall 2027 (Expected)*
Doctor of Philosophy (PhD), Systems Engineering — 5G

Stevens Institute of Technology *2020 – 2024*
Graduate Certificate, Space Systems Engineering

New Jersey Institute of Technology *2014 – 2016*
Master of Science (MS), Electrical Engineering — Wireless Communications

Penn State University *2010 – 2014*
Bachelor of Science (BS), Electrical Engineering

CERTIFICATIONS

Project Management Professional (PMP), May 2022 · Licensed Professional Engineer (PE), New York, Dec. 2019 — Electronics, Controls & Communications · Fundamentals of Engineering (FE), Jan. 2017 · IEEE Wireless Communications Professional (WCP), Oct. 2018

PUBLICATIONS

[1] E. Forbes *et al.*, "Coexistent Evaluation on the Effects of Radar PRF and 5G Symbol-Based Scheduling," *IEEE MILCOM*, Los Angeles, CA, Oct. 2025. **Poster Session Honorable Mention.**

[2] E. Forbes *et al.*, "5G and Radar Coexistence: Demonstrating Pulsed Radar Interference and 5G Performance Metrics," *IEEE MILCOM Demo Session*, Oct. 2025.

[3] E. Forbes *et al.*, "Cellular Data-Based Indoor Localization for Smart Health Applications," *IEEE CHASE*, Jun. 2025.

[4] E. Forbes *et al.*, "Communication Systems for Autonomous eVTOL: Regulatory Challenges, Emerging Technologies, and Future Directions," *IEEE*, 2025.

[5] E. Forbes, E. Silverstein, and Y. Wang, "Elevation-Aware BLER-Centric Link Adaptation for Dynamic LEO 5G NTN Systems," *2026 IEEE Aerospace Conference*, 2026.